

MAMADOU BA

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EDUCATION

City University of New York, College of Staten Island

Expected May 2027

Bachelor of Science in Computer Science

Relevant coursework: Data Structures & Algorithms (C++), Introduction to Data Science (Python), Computer Organization, Discrete Mathematics, Calculus III, Linear Algebra

TECHNICAL SKILLS

- **Languages:** Python, SQL, TypeScript, JavaScript, C/C++, HTML/CSS
- **Data & Analytics:** Pandas, NumPy, scikit-learn, Power BI, ETL/data pipelines, SQL schema design
- **AI/ML & Agents:** Claude API (tools, system prompts), LangChain, RAG pipelines, ChromaDB, evaluation harnesses, Google Gemini API, Microsoft Copilot Studio, Power Automate
- **Tools:** Git/GitHub, Visual Studio Code, Jupyter, Excel

RELEVANT EXPERIENCE

Apex Forum | Hybrid | Operations Analyst & AI Lead

March 2025 – Present

- Partnered with operations staff to scope and prioritize automation projects, then shipped Copilot Studio and Power Automate workflows that cut 8+ hours of weekly manual work on meeting prep and mentorship matching.
- Built Python and Power BI analytics pipelines processing podcast and community engagement data, surfacing insights that drove 30% growth in member participation across 5+ programs.
- Analyzed operational processes to identify where AI automation was and wasn't appropriate, then delivered recommendations to senior leadership that drove a 25% improvement in team efficiency.
- Ran small-group walkthroughs so non-technical staff could maintain and modify AI workflows after handoff, writing runbooks and troubleshooting guides to ensure tools kept running independently.

CSI Innovation Hub x Verizon | Staten Island, NY | Tech Entrepreneurship Mentor

May 2026 – June 2026

- Facilitated weekly sessions for high school students in the Verizon Emerging Tech Impact Program, teaching AI tools, business model design, market research, and product prototyping.
- Mentored Team Purple in designing a Queue Management App for Seniors, guiding them from problem identification through user-centered design and a final presentation at the live Tech Showcase.
- Balanced technical guidance with empathy and encouragement, helping students navigate ethical implications of building technology for seniors and fostering a collaborative team culture.

PROJECTS

VibeFinder – Music Recommender RAG System | Python, LangChain, Claude API, Streamlit | [GitHub](#) | [Live Demo](#)

- Shipped an end-to-end RAG system with Claude API including retrieval pipeline, orchestration logic, configurable scoring modes, and output guardrails validating every model response before delivery.
- Built a 12-check evaluation harness across 3 adversarial profiles to surface failure modes and catch confident-sounding outputs on weak matches, achieving a 100% pass rate before deployment on Streamlit.

The Unofficial Guide – RAG Knowledge Base | Python, ChromaDB, Groq, Gradio | [GitHub](#) | [Live Demo](#)

- Built and deployed a public Hugging Face Space RAG system answering plain-language questions about university courses using 1,700+ real student reviews, with grounded generation, programmatic source attribution, and hard refusal when retrieved context is insufficient.
- Designed a custom chunking pipeline preserving per-review boundaries and cosine-similarity retrieval via ChromaDB and a 5-question eval harness (5/5 accurate answers).

Retail Logistics Data Quality Analysis | Python, Pandas, scikit-learn, Power BI | [GitHub](#)

- Quantified financial impact of poor data quality by analyzing 10,000+ logistics records using anomaly detection algorithms, identifying \$2.3M annual cost impact.
- Developed data governance recommendations based on root-cause analysis of data quality issues, proposing process improvements to optimize supply chain workflows.

LEADERSHIP & VOLUNTEERING

Mouse.org | Remote | Volunteer Mentor & Project Reviewer

Jan 2026 – Jun 2026

- Mentored an NYC special education CS classroom at P721Q JFK Jr. School through the AI League program, sharing career experience and providing feedback on student app prototypes and product pitches.
- Served as project reviewer for the Emoti-Con NYC Youth Tech Expo, evaluating 10 student group projects across health, civic tech, and community categories from NYC middle and high schools.